

AI-ASSISTED THREAT DETECTION DASHBOARD

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AI-Assisted Threat Detection Dashboard

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ABSTRACT

Artificial Intelligence (AI) has become an integral part of modern cybersecurity due to the increasing complexity and frequency of cyber threats. Traditional security systems primarily rely on predefined rules and signatures to identify malicious activities. Although effective against known attacks, these systems often struggle to detect newly emerging threats and sophisticated attack patterns. Machine Learning (ML) techniques overcome these limitations by learning from historical data, recognizing hidden patterns, and identifying suspicious activities with greater accuracy.

This research document presents a comprehensive study of AI-assisted threat detection using machine learning techniques. It examines widely used cybersecurity datasets, explores various machine learning models employed for intrusion detection, and compares their performance based on accuracy, advantages, and limitations. The report also recommends the most suitable model for developing an AI-assisted threat detection dashboard capable of supporting cybersecurity analysts in identifying potential threats efficiently.

Keywords: Artificial Intelligence, Machine Learning, Cybersecurity, Threat Detection, Intrusion Detection, Random Forest, XGBoost, CICIDS2017, Data Visualization.

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1. INTRODUCTION

Cybersecurity has become one of the most important aspects of today's digital society. As organizations increasingly depend on computer networks, cloud platforms, and internet-based applications to perform daily operations, the risk of cyberattacks has also increased significantly. Cybercriminals continuously develop new techniques to exploit system vulnerabilities, steal sensitive information, disrupt business operations, and compromise digital infrastructure. These cyber threats pose serious financial, operational, and reputational risks to individuals, businesses, and governments.

Traditional cybersecurity systems mainly depend on signature-based detection methods and predefined security rules to identify malicious activities. While these approaches are effective in detecting known attacks, they often fail to recognize unknown threats, zero-day attacks, and evolving attack patterns. As a result, organizations require more intelligent and adaptive security solutions capable of learning from data and identifying suspicious activities automatically.

Artificial Intelligence (AI) and Machine Learning (ML) have emerged as powerful technologies for enhancing cybersecurity. Machine learning algorithms analyze historical network traffic, identify patterns associated with malicious behavior, and classify network activities as either normal or suspicious. These intelligent systems improve detection accuracy, reduce false alarms, and assist cybersecurity professionals in responding to incidents more efficiently.

This research focuses on studying AI-assisted threat detection by evaluating benchmark cybersecurity datasets and comparing different machine learning models used for intrusion detection. The objective is to understand the strengths and limitations of various algorithms and identify the most suitable model for building an effective AI-assisted threat detection dashboard.

2. PROBLEM STATEMENT

The rapid growth of digital technologies has significantly increased the number and complexity of cybersecurity threats faced by organizations worldwide. Every day, organizations generate enormous volumes of security logs, network traffic data, and system events. Security analysts often struggle to manually examine this information because of the large number of alerts, many of which are false positives.

Traditional intrusion detection systems primarily rely on predefined rules that cannot effectively identify new attack techniques or previously unseen malicious behavior. Consequently, organizations require intelligent threat detection systems capable of automatically learning attack patterns, identifying abnormal activities, and improving detection accuracy.

Machine Learning provides an effective solution by analyzing network traffic data and recognizing hidden relationships between normal and malicious activities. However, selecting the most suitable machine learning model remains an important challenge because different algorithms perform differently depending on dataset characteristics, feature selection, and computational requirements.

Therefore, this research aims to study and compare multiple machine learning algorithms to determine the most effective model for AI-assisted cybersecurity threat detection.

3. PROJECT OBJECTIVES

The primary objective of this research is to study various Artificial Intelligence and Machine Learning techniques used for cybersecurity threat detection and identify the most suitable model for detecting cyber threats efficiently.

The specific objectives of this research are:

- To understand the concept of AI-assisted cybersecurity threat detection.
- To study benchmark cybersecurity datasets used in intrusion detection research.
- To analyze different machine learning algorithms suitable for cybersecurity applications.
- To compare machine learning models based on accuracy, advantages, and limitations.
- To recommend the most appropriate machine learning model for developing an AI-assisted threat detection dashboard.
- To provide a research foundation for implementing intelligent cybersecurity monitoring systems.

4. ABOUT CYBERSECURITY THREAT DETECTION

Cybersecurity threat detection is the process of identifying malicious activities, suspicious behavior, and security breaches within computer systems and networks before they cause significant damage. The primary objective of threat detection is to recognize potential attacks at an early stage so that appropriate preventive actions can be taken to protect organizational resources.

Modern organizations generate massive amounts of network traffic, system logs, and user activity records every day. Manually monitoring such a large volume of information is difficult and time-consuming. Therefore, Artificial Intelligence (AI) and Machine Learning (ML) have become essential technologies for automating threat detection and improving cybersecurity operations.

Unlike traditional security systems that rely on predefined rules, AI-based systems learn patterns from historical data and continuously improve their ability to detect malicious activities. These systems can identify both known and unknown attacks by recognizing unusual behavior and network anomalies. Machine learning algorithms analyze multiple features such as network traffic, packet information, communication protocols, connection duration, and user behavior to determine whether an activity is normal or malicious.

An AI-assisted threat detection system not only improves detection accuracy but also reduces false alarms, shortens incident response time, and helps cybersecurity analysts prioritize critical threats. As cyberattacks continue to evolve, intelligent threat detection systems have become an important component of modern cybersecurity infrastructures.

Common cybersecurity threats include:

- Malware
- Phishing Attacks
- Ransomware
- Denial of Service (DoS)
- Distributed Denial of Service (DDoS)
- Brute Force Attacks
- SQL Injection
- Cross-Site Scripting (XSS)
- Insider Threats
- Botnet Attacks

These attacks vary in complexity and impact, making intelligent detection systems essential for protecting digital assets.

5. RESEARCH ON CYBERSECURITY DATASETS

The success of any machine learning model depends heavily on the quality of the dataset used for training and evaluation. A well-designed cybersecurity dataset contains realistic network traffic, multiple attack categories, and sufficient features that enable machine learning algorithms to distinguish between normal and malicious behavior. Several benchmark datasets have been developed by researchers to evaluate intrusion detection systems. Among these, CICIDS2017, NSL-KDD, and UNSW-NB15 are the most widely used datasets in cybersecurity research.

5.1 CICIDS2017 Dataset

The CICIDS2017 dataset was developed by the Canadian Institute for Cybersecurity (CIC). It is one of the most comprehensive and realistic datasets available for intrusion detection research. The dataset contains normal network traffic along with various categories of cyberattacks captured over multiple

days. It closely represents real-world network environments and includes more than eighty network traffic features.

Researchers prefer the CICIDS2017 dataset because it includes modern attack scenarios such as brute force attacks, denial-of-service attacks, distributed denial-of-service attacks, web attacks, botnet activities, infiltration attacks, and port scanning. The availability of detailed traffic features allows machine learning algorithms to achieve high detection accuracy.

Dataset Overview

Property	Description
Dataset Name	CICIDS2017
Developed By	Canadian Institute for Cybersecurity (CIC)
Release Year	2017
File Format	CSV
Data Type	Network Flow Records
Number of Features	More than 80
Contains Normal Traffic	Yes
Contains Attack Traffic	Yes
Purpose	Intrusion Detection Research

5.2 NSL-KDD Dataset

The NSL-KDD dataset is an improved version of the older KDD Cup 1999 dataset. It was created to eliminate duplicate records and improve the evaluation of intrusion detection systems. Although it is an older dataset, it is still widely used for benchmarking machine learning algorithms because of its balanced structure and labeled attack categories.

The dataset includes various attack types such as Denial of Service (DoS), Probe attacks, User-to-Root (U2R), and Remote-to-Local (R2L) attacks. It provides a good foundation for comparing classical machine learning algorithms.

Dataset Overview

Property	Description
Dataset Name	NSL-KDD
Release Year	2009
File Format	CSV
Purpose	Intrusion Detection
Contains Labeled Data	Yes
Common Usage	Machine Learning Benchmark

5.3 UNSW-NB15 Dataset

The UNSW-NB15 dataset was developed by the Australian Centre for Cyber Security to overcome the limitations of older intrusion detection datasets. It contains modern network traffic generated in a realistic laboratory environment and includes both normal activities and multiple categories of cyberattacks.

The dataset provides forty-nine network traffic features describing each connection. It contains attacks such as Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic attacks, Reconnaissance, Shellcode, and Worms.

The UNSW-NB15 dataset is considered highly suitable for evaluating advanced machine learning algorithms because it reflects contemporary cybersecurity threats more accurately than older datasets.

Dataset Overview

Property	Description
Dataset Name	UNSW-NB15
Developed By	Australian Centre for Cyber Security
Release Year	2015
File Format	CSV
Number of Features	49
Purpose	Modern Intrusion Detection

6. DATASET COMPARISON

Different cybersecurity datasets have different characteristics and are suitable for different research objectives. Comparing benchmark datasets helps researchers choose the most appropriate dataset for training machine learning models.

Dataset	Advantages	Limitations	Suitability
CICIDS2017	Modern attacks, realistic traffic, large number of features	Large file size	Excellent
NSL-KDD	Clean dataset, balanced records, easy to understand	Older attack types	Good
UNSW-NB15	Modern attack scenarios, realistic environment	Slightly complex preprocessing	Very Good

Among these datasets, **CICIDS2017** is selected for this research because it contains realistic network traffic, diverse cyberattack categories, and detailed network flow features that improve the performance of machine learning algorithms.

7. IMPORTANT FEATURES OF THE CICIDS2017 DATASET

Machine learning models use various network traffic features to classify activities as normal or malicious. Some of the most important features available in the CICIDS2017 dataset are listed below.

Feature	Description
Source IP	IP address of the sender
Destination IP	IP address of the receiver
Source Port	Sender port number
Destination Port	Receiver port number

Protocol	Communication protocol used
Flow Duration	Total duration of the network connection
Total Forward Packets	Packets sent from source to destination
Total Backward Packets	Packets returned from destination
Packet Length Mean	Average packet size
Flow Bytes per Second	Data transfer rate
Flow Packets per Second	Number of packets transmitted each second
Label	Indicates Normal or Attack traffic

These features provide sufficient information for machine learning algorithms to identify suspicious behavior, detect anomalies, and classify cyberattacks accurately. By analyzing patterns within these network characteristics, AI models can distinguish between legitimate network communication and malicious activities with high accuracy.

8. MACHINE LEARNING MODELS FOR THREAT DETECTION

Machine Learning (ML) plays a significant role in modern cybersecurity by automatically identifying malicious activities from large volumes of network traffic data. Different machine learning algorithms have different strengths, computational requirements, and prediction capabilities. Selecting an appropriate model is essential for developing an effective AI-assisted threat detection system. The following machine learning models are commonly used for cybersecurity threat detection.

8.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm primarily used for binary classification problems. It predicts the probability that a network activity belongs to a particular category, such as normal traffic or malicious traffic. Although Logistic Regression is simple and computationally efficient, its performance decreases when dealing with highly complex and non-linear cybersecurity datasets.

Advantages

- Simple and easy to implement.
- Fast training and prediction.
- Easy to interpret results.
- Suitable for baseline classification.

Limitations

- Lower accuracy for complex datasets.
 - Cannot effectively model non-linear relationships.
 - Sensitive to correlated features.
-

8.2 Decision Tree

Decision Tree is a supervised machine learning algorithm that classifies data using a hierarchical tree structure. Each internal node represents a decision based on a feature, while each leaf node represents the final prediction. Decision Trees are widely used because they are easy to understand and require minimal data preprocessing.

Advantages

- Easy to visualize and interpret.
- Handles numerical and categorical data.
- Requires minimal preprocessing.
- Fast prediction.

Limitations

- Can easily overfit training data.

- Less stable when dataset changes.
 - Lower generalization capability compared to ensemble methods.
-

8.3 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees to improve prediction accuracy and reduce overfitting. Each decision tree independently predicts the output, and the final prediction is determined through majority voting. Random Forest has become one of the most popular machine learning algorithms for cybersecurity because of its robustness and excellent classification performance.

Advantages

- High detection accuracy.
- Reduces overfitting.
- Handles large datasets efficiently.
- Works well with high-dimensional data.
- Robust against noisy data.

Limitations

- Higher computational cost.
 - Requires more memory.
 - Less interpretable than a single Decision Tree.
-

8.4 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification by identifying the optimal boundary that separates different classes. In cybersecurity, SVM effectively distinguishes between normal and malicious network traffic using mathematical optimization techniques.

Advantages

- High classification accuracy.

- Effective for high-dimensional datasets.
- Strong mathematical foundation.
- Performs well on moderate-sized datasets.

Limitations

- Slow training on very large datasets.
 - Requires parameter tuning.
 - Computationally expensive.
-

8.5 XGBoost (Extreme Gradient Boosting)

XGBoost is an advanced ensemble learning algorithm that builds multiple decision trees sequentially using gradient boosting techniques. It has become one of the highest-performing machine learning algorithms for structured datasets and cybersecurity research due to its exceptional prediction accuracy and scalability.

Advantages

- Excellent prediction accuracy.
- Handles missing values effectively.
- Fast training using parallel processing.
- Performs well on structured cybersecurity datasets.
- Reduces overfitting through regularization.

Limitations

- More complex implementation.
 - Requires careful hyperparameter tuning.
 - Higher computational requirements.
-

8.6 Isolation Forest

Isolation Forest is an unsupervised anomaly detection algorithm designed to identify unusual observations by isolating them from normal data. Since cyberattacks often appear as anomalies within network traffic, Isolation Forest is widely used for detecting unknown threats without requiring labeled datasets.

Advantages

- Excellent for anomaly detection.
- Does not require labeled attack data.
- Fast execution on large datasets.
- Detects previously unseen attacks.

Limitations

- Less suitable for multi-class classification.
- May generate false positives.
- Performance depends on anomaly distribution.

9. COMPARISON OF MACHINE LEARNING MODELS

The performance of machine learning algorithms varies depending on the characteristics of the dataset, feature engineering techniques, and model configuration. The following table presents a comparative analysis of commonly used machine learning models for cybersecurity threat detection based on representative benchmark results reported in research literature.

Machine Learning Model	Accuracy (%)	Advantages	Limitations
Logistic Regression	89	Simple, Fast, Easy to Interpret	Lower Accuracy
Decision Tree	94	Easy to Understand, Fast	Overfitting
Random Forest	99	High Accuracy, Robust	Higher Memory Usage

Support Vector Machine	97	Excellent Classification	Computationally Expensive
XGBoost	99.3	Highest Accuracy, Scalable	Complex Parameter Tuning
Isolation Forest	92	Excellent Anomaly Detection	Higher False Positives

Comparative Analysis

The comparison indicates that ensemble learning algorithms consistently outperform traditional classification models for cybersecurity applications. Random Forest and XGBoost achieve the highest detection accuracy because they combine multiple decision trees to improve prediction performance while reducing overfitting.

Support Vector Machine also performs well but requires greater computational resources. Logistic Regression and Decision Tree provide simple baseline models but generally produce lower accuracy when applied to complex cybersecurity datasets.

Isolation Forest differs from other models because it focuses on anomaly detection rather than supervised classification. It is particularly effective for identifying previously unseen attacks and abnormal network behavior.

Considering detection accuracy, computational efficiency, scalability, and practical implementation, Random Forest and XGBoost emerge as the most suitable algorithms for AI-assisted cybersecurity threat detection.

Note: The accuracy values presented above are representative benchmark values compiled from published cybersecurity research studies using datasets such as CICIDS2017, NSL-KDD, and UNSW-NB15. Actual model performance may vary depending on data preprocessing, feature engineering, feature selection, and hyperparameter optimization.

10. PROPOSED AI-ASSISTED THREAT DETECTION WORKFLOW

An AI-assisted threat detection system follows a structured workflow that transforms raw network traffic into meaningful security insights. The process begins with collecting network traffic data from various sources, followed by preprocessing and feature extraction. The processed data is then analyzed using machine learning algorithms to classify activities as either normal or malicious. Finally, the detected threats are displayed through a dashboard that enables security analysts to monitor incidents and respond quickly.

The proposed workflow consists of the following stages:

1. Network Traffic Collection
2. Data Preprocessing
3. Feature Extraction
4. Machine Learning Model
5. Threat Classification
6. Dashboard Visualization
7. Security Response

Workflow Description

Network Traffic Collection: Data is collected from firewalls, routers, switches, servers, and network monitoring tools.

Data Preprocessing: Missing values are handled, duplicate records are removed, categorical features are encoded, and numerical values are normalized.

Feature Extraction: Important network characteristics such as protocol type, packet size, flow duration, source IP, destination IP, and packet count are extracted for analysis.

Machine Learning Model: The processed data is analyzed using machine learning algorithms such as Random Forest, XGBoost, Support Vector Machine, Logistic Regression, Decision Tree, and Isolation Forest.

Threat Classification: The trained model classifies each network flow as either normal traffic or a specific cyberattack.

Dashboard Visualization: The classification results are displayed through an interactive dashboard using charts, graphs, and summary statistics to support decision-making.

Security Response: Security analysts investigate detected threats and initiate appropriate mitigation actions.

Network Traffic



Data Collection



Data Preprocessing



Feature Extraction



Machine Learning Model



Threat Detection



Dashboard

11. BEST MODEL SELECTION

Selecting the most suitable machine learning model is an important step in developing an AI-assisted threat detection system. Although several algorithms demonstrate good performance, each model has different strengths depending on the complexity of the dataset, computational requirements, and detection objectives.

Based on the comparative analysis conducted in this research, **Random Forest** is selected as the recommended machine learning model. Random Forest consistently achieves high detection accuracy while minimizing overfitting through ensemble learning techniques. It effectively handles high-dimensional cybersecurity datasets and performs well even when some features contain noise or missing values.

XGBoost also demonstrates outstanding performance and achieves the highest reported accuracy in many benchmark studies. However, it requires careful parameter tuning and greater computational resources, making it slightly more complex to implement than Random Forest.

Support Vector Machine provides excellent classification performance but becomes computationally expensive for large-scale cybersecurity datasets. Logistic Regression and Decision Tree serve as useful baseline models but generally achieve lower prediction accuracy for modern intrusion detection tasks.

Isolation Forest is particularly useful for anomaly detection because it can identify suspicious network activities without requiring labeled attack data. However, it is better suited for detecting anomalies than performing multi-class attack classification.

Final Recommendation

Evaluation Criteria	Recommended Model
Overall Performance	Random Forest
Average Accuracy	Approximately 99%
Ease of Implementation	High
Scalability	Excellent
Overfitting Resistance	High

Large Dataset Support	Excellent
Recommended for Threat Detection	Yes

Based on these evaluation criteria, Random Forest provides the best balance between prediction accuracy, computational efficiency, scalability, and implementation simplicity. Therefore, it is recommended as the primary machine learning algorithm for developing an AI-assisted threat detection dashboard.

12. FUTURE ENHANCEMENTS

Artificial Intelligence continues to evolve rapidly, creating new opportunities for improving cybersecurity systems. Future AI-assisted threat detection platforms can become more intelligent, scalable, and automated by integrating advanced technologies and real-time threat intelligence.

Several enhancements can be incorporated into future versions of the proposed system:

- Integration with cloud security platforms such as AWS, Microsoft Azure, and Google Cloud.
- Real-time monitoring of live network traffic instead of offline datasets.
- Integration with Threat Intelligence Platforms such as VirusTotal, AbuseIPDB, and AlienVault Open Threat Exchange.
- Adoption of Explainable Artificial Intelligence (XAI) to improve transparency and explain model predictions.
- Use of Deep Learning models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformer-based architectures for detecting advanced cyberattacks.
- Deployment using cloud-native technologies for scalable threat detection.
- Development of interactive dashboards with automated alert generation and incident reporting.
- Continuous model retraining using newly collected cybersecurity data to improve detection performance against emerging threats.

These improvements would enhance detection accuracy, reduce response time, and strengthen organizational cybersecurity resilience.

13. CONCLUSION

Artificial Intelligence has become one of the most promising technologies for strengthening modern cybersecurity systems. Traditional signature-based detection techniques are no longer sufficient to identify rapidly evolving cyber threats. Machine Learning provides intelligent mechanisms capable of learning from historical network traffic, identifying abnormal behavior, and detecting malicious activities with greater accuracy.

This research examined benchmark cybersecurity datasets including CICIDS2017, NSL-KDD, and UNSW-NB15, with particular emphasis on the CICIDS2017 dataset due to its realistic network traffic and comprehensive attack categories. Multiple machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, XGBoost, and Isolation Forest were studied and compared based on their working principles, advantages, limitations, and reported benchmark performance.

The comparative analysis demonstrated that ensemble learning algorithms outperform traditional classification techniques for cybersecurity threat detection. Random Forest provides an excellent balance between prediction accuracy, robustness, computational efficiency, and ease of implementation, while XGBoost achieves the highest reported benchmark accuracy for many intrusion detection tasks.

Overall, this research concludes that Artificial Intelligence significantly improves cybersecurity threat detection by enabling automated analysis of network traffic, reducing false alarms, and supporting rapid incident response. The findings presented in this report provide a strong foundation for developing an AI-assisted threat detection dashboard capable of assisting cybersecurity professionals in monitoring and protecting modern digital infrastructures.

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